Real-Time Closed-Loop Brain-Body Simulation Using a GPU-Based Multi-Region Spiking Neural Network and a Mouse Musculoskeletal Model

GPUベース多領域 SNNとマウス筋骨格モデルを使ったリアルタイム閉ループ脳身体シミュレーション

Masachika Taniguchi, Michiru Inoue, Tadashi Yamazaki

Graduate School of Informatics and Engineering, The University of Electro-Communications

GPU-based SNN ran the brain–body loop faster than real time overall.

10.05 s simulated in 9.38 s; 50-ms blocks averaged 46.6 ms.

INTRO

- Motor control emerges from closed-loop interactions between brain, body, and sensory feedback.
- Here, we coupled a GPU-based multi-region SNN with a mouse musculoskeletal forearm model to build a real-time brain–body loop.

METHODS

- A multi-region SNN including M1/S1, BG, TH, cerebellum, and CPG was simulated on a GPU.
- M1_L5B_PT activity was converted to antagonistic extensor/flexor muscle activation.
- Body movement was fed back to S1: M1→muscles→body→S1

RESULTS

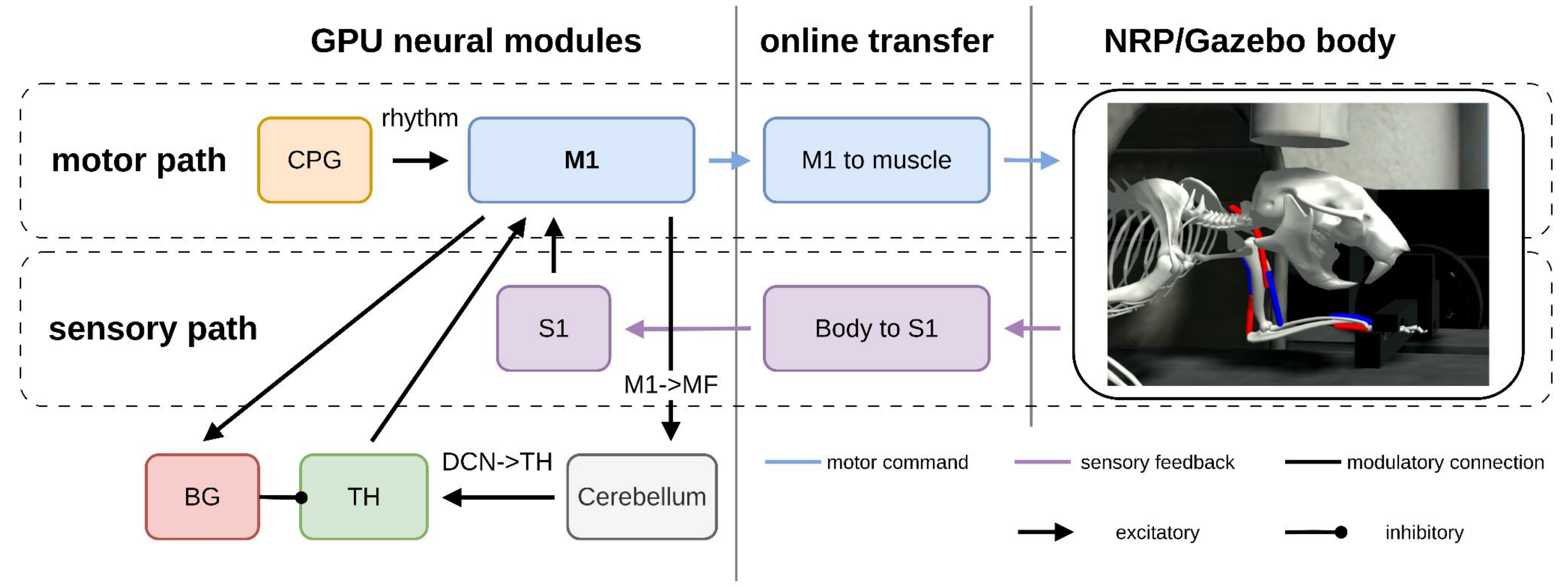
- Alternating M1 output generated antagonistic muscle activity.
- The forearm and lever showed periodic push–pull movement.
- S1 activity changed with body movement, consistent with body-derived sensory feedback.

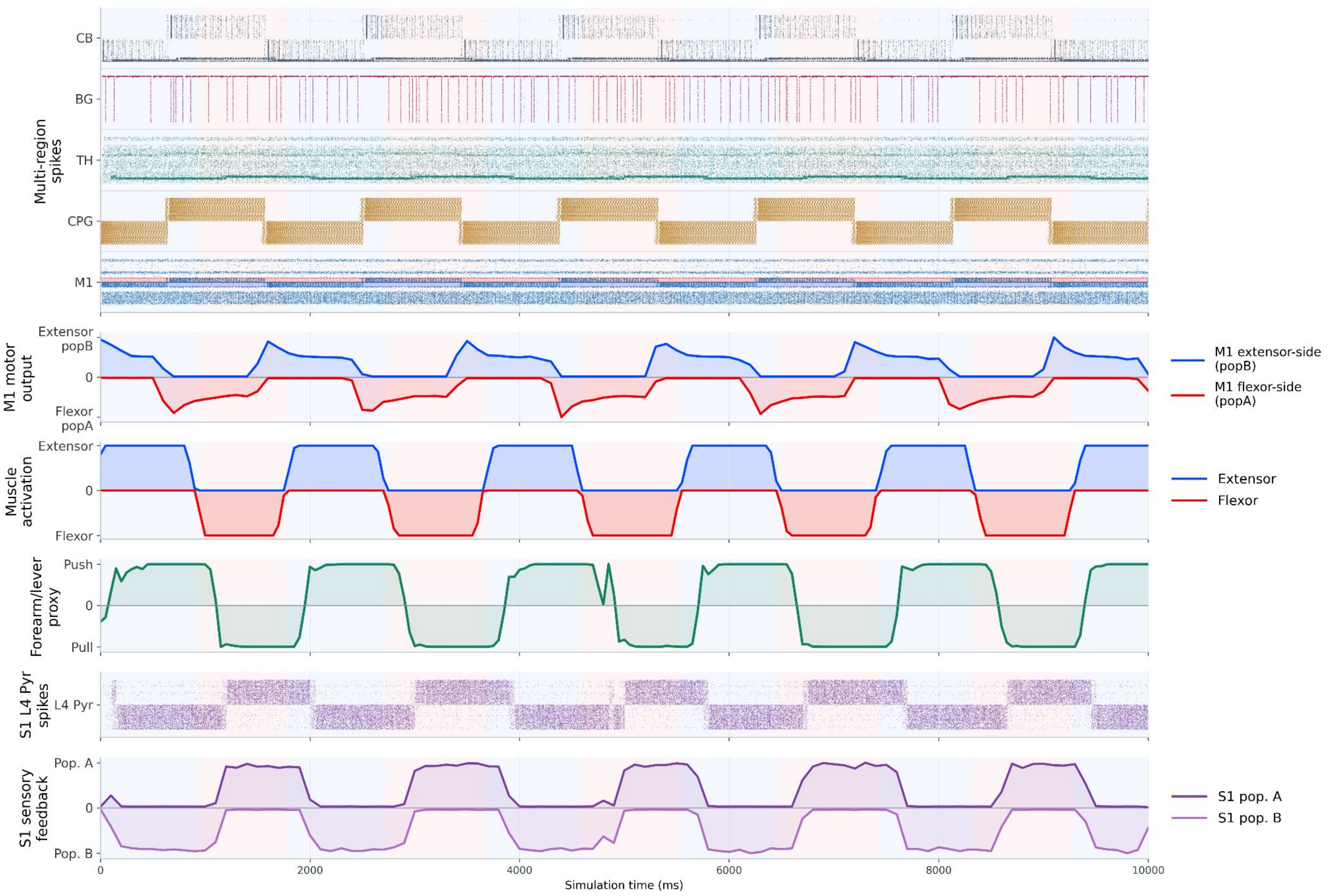
CONCLUSION

- GPU execution enabled an overall real-time brain–body loop.
- The loop generated M1-driven antagonistic muscle activity and push–pull movement.
- This framework can test how sensory feedback and cerebellar modulation shape movement.

REFERENCES

1. Kuniyoshi et al., Front. Neurorobot., 2023.
2. Falotico et al., Front. Neurorobot., 2017.
3. Kuriyama et al., Front. Cell. Neurosci., 2021.





Simulation / wall-clock
1.07x
10.05 s simulated / 9.38 s wall

Loop latency
46.6 ms
mean; target = 50 ms per block

Blocks within target
74%
149/201 blocks <= 50 ms; max 63.6 ms

GPU utilization
79.2%
Tesla V100 active loop; 2.52 GiB